Machine Learning : The No-Nonsense (But Fun) Guide - Day 2

Linear Regression: All you need to know !!

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1. What Is the Goal of Linear Regression?

Imagine you have a collection of data points and want to discover the relationship between them.

Linear regression helps you draw the **best-fitting straight line** through those points to predict future values.

Example

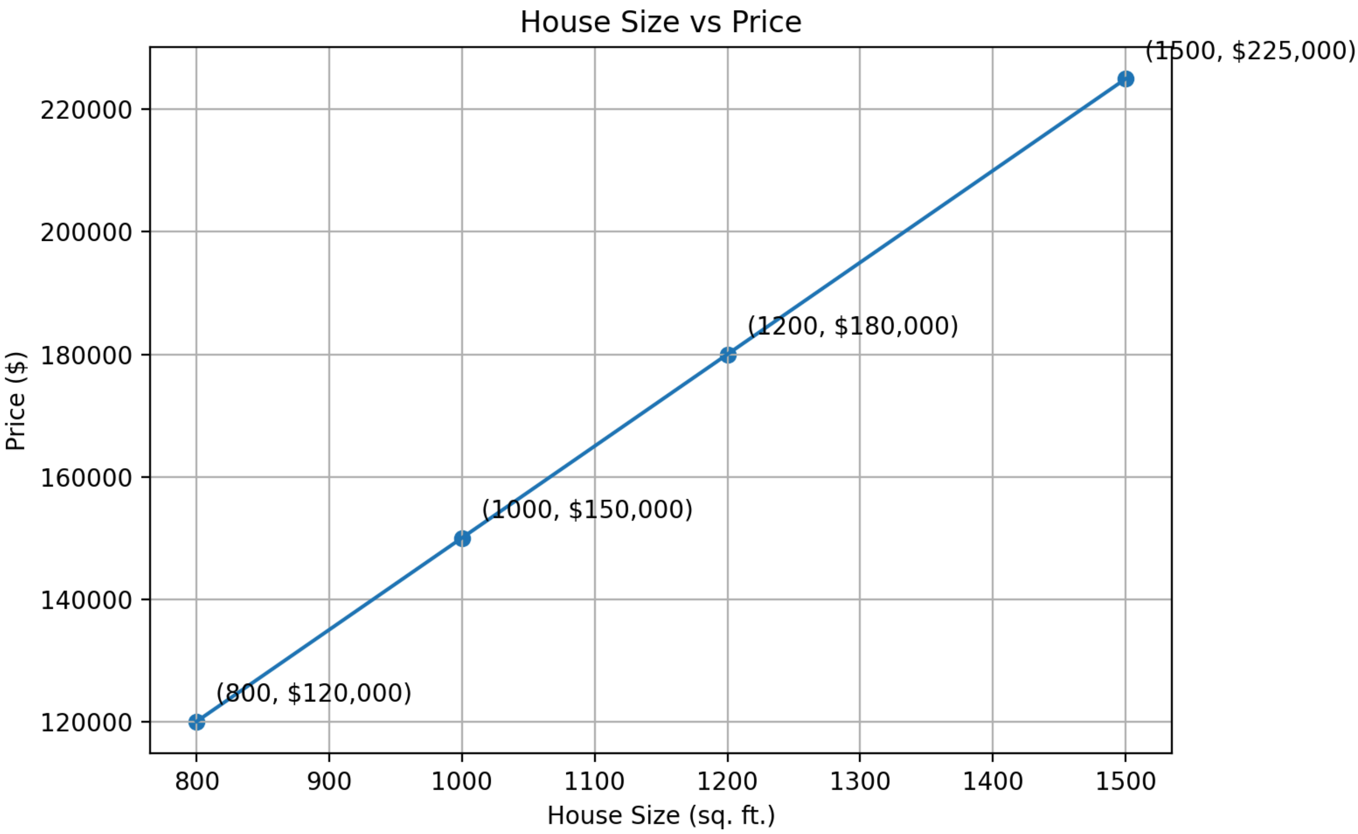
Suppose you have information about house sizes and their prices.

House Size (sq. ft.)	Price ($)
800	120,000
1000	150,000
1200	180,000
1500	225,000

Linear regression finds a line that helps answer questions like:

> "If a house is 1,300 sq. ft., what should its price be?"

Visual Representation



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2. The Mathematical Formula

Simple Linear Regression

[
y = mx + b
]

Machine Learning Notation

[
y(hat) = wx + b
]

Where:

Symbol	Meaning	Description
x	Feature	Input value

y	Actual value	True output
ŷ	Predicted value	Model output
w or m	Weight/Coefficient	Importance of the feature
b	Bias/Intercept	Base value

Example

House Price = Size Influence * Size + Base Price

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3. Multiple Linear Regression

When multiple features affect the output, we use multiple linear regression.

Example

House Price = Influence of Size

- Influence of Number of Rooms
- Influence of Location
- (Base Price

Feature Table

Feature	Variable	Influence
Size	(x_1)	(w_1)
Number of Rooms	(x_2)	(w_2)

Location Score	(x_3)	(w_3)
Base Price	—	(b)

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4. How Does the Model Learn?

The learning process involves finding the best values for the weights and bias.

Training Steps

1. Start with random weights and bias.
2. Make predictions.
3. Calculate the error.
4. Adjust weights to reduce error.
5. Repeat until the error becomes minimal.

Learning Workflow

Initialize Parameters

↓

Make Predictions

↓

Calculate Error

↓

Update Parameters

↓

Repeat Process

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5. Measuring Error: Mean Squared Error (MSE)

The model needs a way to measure how wrong its predictions are.

Where:

Symbol	Meaning
n	Number of observations
y	Actual value
ŷ	Predicted value

Interpretation

- Lower MSE = Better model
- Higher MSE = Poor predictions

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6. Improving the Model: Gradient Descent

Gradient descent is an optimization algorithm used to reduce prediction error.

Imagine This Scenario

You are standing on a hill with a blindfold and want to reach the lowest point.

You cannot see the path, so you:

- Feel the slope around you.
- Take a step downhill.
- Repeat until the ground becomes flat.

This is exactly how gradient descent works.

7. Understanding the Learning Rate

The learning rate determines the size of each step taken during gradient descent.

Choosing the Right Learning Rate

Learning Rate	Result
Too Small	Learning is very slow

Too Large	May overshoot the minimum
Just Right	Converges efficiently

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8. Applications of Linear Regression

Linear regression works best when predicting continuous numerical values.

Common Applications

- House price prediction
- Sales forecasting
- Temperature prediction
- Demand forecasting
- Revenue estimation
- Stock trend analysis
- Risk assessment

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9. When Should You Use Linear Regression?

Suitable Situations

Condition	Suitable
Relationship is approximately linear	✓
Output is numerical	✓
Model should be easy to interpret	✓
Fast training is required	✓
Dataset has low noise	✓

Example

Advertising Budget → Sales Revenue

Study Hours → Exam Scores

House Size → House Price

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10. When Should You Avoid Linear Regression?

Unsuitable Situations

Condition	Suitable?
Predicting categories (Spam/Not Spam)	✗
Complex non-linear relationships	✗
Large number of interacting features	✗
Presence of many outliers	✗
Strong feature correlation	✗

Example

Email → Spam or Not Spam

Use Classification Algorithms instead.

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11. Assumptions of Linear Regression

For best performance, linear regression assumes:

1. Linear relationship between input and output.

2. Independent observations.
3. Minimal multicollinearity among features.
4. Constant variance of errors (homoscedasticity).
5. Errors are approximately normally distributed.
6. Few or no significant outliers.

Linear Regression Formula Cheat Sheet

1. Simple Linear Regression

$\hat{y} = wx + b$

Predicted Value = (Weight × Feature) + Bias

2. Multiple Linear Regression

$\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 + b$

Prediction = Sum of feature influences + Bias

3. Mean Squared Error (MSE)

$MSE = (1/n)\ \Sigma(y - \hat{y})^2$

Measures the average prediction error

4. Gradient Descent Update

$w = w - \alpha \times \partial J/\partial w$

New Weight = Old Weight - (Learning Rate × Slope)

5. Example: House Price Prediction

House Price = (150 × Size) + 0

For a 1300 sq. ft. house: Price = 150 × 1300 = $195,000

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12. Interviews Questions based on Linear Regression.

Most Frequently Asked Questions (Top 10)

1. What are the assumptions of linear regression?
2. What is multicollinearity, and how do you detect it?
3. What is the difference between Ridge and Lasso regression?
4. What is R-squared and Adjusted R-squared?
5. How do you handle categorical variables in linear regression?
6. What is gradient descent?

7. How do you identify overfitting?
8. What are residual plots used for?
9. How do you handle missing values and outliers?
10. What is the bias-variance trade-off?

Thank you for reading. Please practice the coding part. Happy Learning !!